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Original article

Energy poverty alleviation through improvement of access to electricity in Nsanke and neighboring villages: A road towards rural electrification projects for sustainable development

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ABSTRACT

Access to energy remains a global problem. In Sub-Saharan Africa (SSA), remote rural areas have multiple sources of energy. Nevertheless, the population suffers from poor energy quality and access. This manuscript's primary strength lies in its thorough empirical analysis of the link between energy access and poverty reduction, particularly in a unique and under-explored region. The research methodology consists of data collection, analysis and interpretation of results, and discussion. We chose to use the probability sampling method with a semi-closed questionnaire. From the door-to-door survey, data were collected on demographic and socio-economic characteristics of households, basic social services, development infrastructure, and lighting facilities. Out of 100 households selected at the beginning of the study, 68 were interviewed due to field constraints and were designed to be representative at the Nsanke village level. Much other information was gathered from field observations. To locate the study area, we used a Garmin handheld GPS receiver and various software, including Mapsource version 8.5 and ArcGIS version 10.5. By applying both descriptive and multivariate statistical methods to primary household data from Nsanke village, we are uncovering crucial information about how access to energy—or the lack of it—contributes to poverty. Our results indicate that up to 91.93% of the households have a total monthly consumption that does not exceed $25, and to meet their needs, someone invests in small, non-farm businesses. 79.03% of the households use kerosene lamps as the only means of lighting. The other energy sources, including generators, battery torches, and solar lighting, are not used systematically but alternate with kerosene lamps. In Nsanke village, cooking is mainly done with firewood as the primary resource under a three-stone hearth. Nsanke and neighboring villages do not have access to clean and affordable water for drinking and cooking. Each of our explanatory variables for lack of access to energy has an internal probability level of 5% with positive and significant odds ratio indices indicating their relationship with the explained variables. Gender (0.252), age (0.263), education of the head of household (0.034), and household size (1.120) have a positive effect on the probability of respondents using kerosene lamps as the only means of lighting in Nsanke village. Variables representing unspecified factors are positive and significant at the 1% threshold, implying that socio-economic factors as determinants of poverty can predict lack of access to energy. In addition, the Chi-square statistic, which confirms the specification of the model, is significant at the 5% level. This in-depth evidence underscores the vital role of electricity in improving living standards and strengthens the arguments for targeted rural electrification initiatives. These findings are highly valuable for policymakers,

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development experts, and researchers focused on energy poverty and rural development, offering essential, data-driven perspectives for future interventions.

Symbols and abbreviations

Y	Dependent variable associated with the event occurring for each individual in the sample,
ε	Error term which is a vector of all unobserved variables
β_i	Estimates of the regression coefficients of the order i
X_i	Independents variables of the order i
n	Number of individuals
ASAIE	African Scientific Association for Innovative and Entrepreneurship
BLR	Binary Logistic Regression
CASCI	Central African Sustainable Cities Initiative
DS	Descriptive Statistics
GCT	Granger Causality Test
IRENA	International Renewable Energy Agency
LEC	*Laboratoire Energétique Carnot*
LRM	Logistic Regression Model
MDGs	Millennium Development Goals
OLS	Multiple Linear Regression
PAPI	Paper And Pencil Interviewing
PCR-TASD	Promotion Centre of Research for Technological Advancement and Sustainable Development
PM	Probit Model
SDGs	Sustainable Development Goals
SE4All	Sustainable Energy for All
SSA	Sub-Saharan Africa

Introduction

Poverty is a very complex concept. Conventional approaches define poverty as low income or low consumption [1]. For a better understanding of various issues related to poverty reduction, other approaches have been considered, emphasizing the multidimensional nature of poverty, including basic needs and capabilities approaches [2–5]. The problem of poverty is particularly acute in Africa, where there is a strong link between poverty and lack of access to an adequate standard of living [6]. For millions of people, a better life means having access to basic needs such as education, health care, food availability [7,8], clothing and housing [9], and access to potable and affordable water [6,10]. None of these needs can be met without energy. Energy is therefore one of the most essential inputs to support people's livelihoods at the most basic level. It is a prerequisite for cooking food, boiling water, and heating [6,10,11]. Indeed, energy has been recognized as a vital factor of production used to facilitate socio-economic well-being [12,13] .

Many authors have analyzed the relationship between energy and household poverty using socio-economic factors as determinants of poverty from different estimation models. The models include Descriptive Statistics (DS), Granger Causality Test (GCT) [14,15], Logistic Regression Model (LRM) [16–18], and Probit Model (PM) [19], among others. The empirical literature shows that household access to energy has a significant impact on socio-economic variables [17,20]. The authors of Refs [17,21] have found that reducing energy poverty helps to reduce income poverty. Using the DS and the PM, Demurger et al. [19] found that income is a key factor in explaining energy consumption and alternative fuels. Using the GCT, Antai et al. [22] found that there is a proportional and significant relationship between energy access and income generation. In general, poverty levels increase when energy is not accessible and affordable [23]. The use of traditional energy sources and the negative impacts associated with the unavailability of modern energy exacerbate poverty by reducing its capacity [24]. Lack of access to clean and affordable energy could then be seen as a fundamental dimension of poverty [25–27].

Under these circumstances, alleviating energy poverty has becomes a critical global energy policy concern. As early as the 1990 s, various interventions were initiated to improve access to electricity, implement renewable energy programs, and develop energy markets [28–30]. In September 2000, world leaders adopted the United Nations Millennium Declaration, committing their nations to a global partnership to eradicate extreme poverty and setting the Millennium Development Goals (MDGs) to be achieved by the end of 2015. In 2015, all United Nations member states adopted the 2030 Agenda for Sustainable Development, a shared blueprint for peace and prosperity for people and the planet, now and in the future. At its heart are 17 Sustainable Development Goals (SDGs), which are an urgent call to action for all countries in global partnership. Other important actions have been taken around the world to eradicate energy poverty. These include the work of Power Africa, the Sustainable Energy for All (SE4All) initiative, and the China South-South Climate Cooperation Fund.

Despite significant improvements in access to electricity in recent decades, the majority of the population in several SSA countries remains unconnected [31]. Recent reports on energy access indicate that two out of three people in SSA do not currently have access to electricity [32], while the remaining one-third cannot consume as much as they would like due to regular power cuts and brownouts caused by structural problems in the electrical systems [33]. The lack of access to electricity in SSA is even more dramatic in rural areas, where electrification rates average 16 %, compared to 99 % in North African countries and 71 % in South Africa [33], and nearly 900 million people in SSA rely on traditional biomass for cooking and heating [34].

In Cameroon, approximately 80–90 % of the population relies on traditional fuels such as wood, charcoal, agricultural residues, and kerosene for cooking, heating, and lighting [11], with potential impacts that include environmental [35–37], social, and economic consequences. Meanwhile, the heavy reliance on traditional fuels in global energy production and consumption has emerged as a major obstacle to achieving sustainable development goals [38–40]. Despite its vast natural resources, energy demand in Cameroon remains unsatisfied, and access to modern energy is very low, with a national average of 15 % for electricity and 18 % for domestic gas [41,42]. In addition, access to electricity is less than 10 % in rural areas, compared to about 50 % in urban areas [42]. The contrast between the country's potential and current energy access (despite high energy demand) can be explained by many challenges, including cost and the poverty level of the population, especially in remote and isolated areas [43]. According to the International Renewable Energy Agency (IRENA), the goals associated with the United Nations SDGs agenda may remain unrealized if the global electrification rates are not significantly improved. This premise motivated the organization African Scientific Association for Innovation and Entrepreneurship (ASAIE) in partnership with *Laboratoire Energétique Carnot* (LEC), Promotion Centre of Research for Technological Advancement and Sustainable Development (PCR-TASD), and PROSOFOR AFRIQUE to initiate the Central African Sustainable Cities Initiative (CASCI), which aims to combat rural–urban migration and stimulate local economies in the Central African region. The outcomes of the

CASCI program can help Central African countries to achieve the SDG target of maximum electrification. The program starts in the village of Nsanke, one of the ten (10) villages of the Mbo'o community in the coastal part of Cameroon, through the present research work. The scientific objective of this research is to critically analyze the role of energy access in poverty alleviation in Nsanke and neighboring villages, highlighting the profound impact that access to electricity has on the socio-economic development of rural communities.

Here we examine the relationship between energy access and rural poverty in order to propose an appropriate solution for improving the living conditions and sustainable development of the inhabitants of Nsanke and neighboring villages. Our results show that the population of Nsanke and neighboring villages is very poor. About 91.93 % of the households have only $25 as their total monthly consumption, and in order to meet their needs, someone invests in small, non-agricultural businesses. 79.03 % of the households use kerosene lamps as their only means of lighting. The other energy sources found in this community (32.3 % for solar torches, 19.35 % for solar lamps, and 8.06 % for solar panels) are not used systematically but alternate with kerosene lamps. In Nsanke village, cooking is mainly done with firewood as the primary resource under a three-stone hearth. Nsanke and neighboring villages do not have access to clean and affordable water for drinking and cooking. Gender, age, education of the head of household, and household size are determinants of energy poverty in Nsanke and neighboring villages. These findings are consistent with some existing literature from some low-income countries, probably Cameroon [44–47]. In line with the conclusion of the authors of Refs [9,48], these findings support our main finding that lack of access to energy is the main cause of poverty for the residents of Nsanke and neighboring villages.

This article is our contribution to improving the overall living conditions of not only the people of Nsanke and neighboring villages but also those in other rural areas around the world. It's also a valuable contribution to the academic literature on sustainable energy. Through a detailed empirical investigation using data from 68 households, it seeks to establish a quantifiable relationship between energy poverty and overall poverty alleviation, highlighting the current reliance on traditional fuels and the low adoption rates of renewable energy sources. By proposing targeted rural electrification projects as a means of improving livelihoods, the study advocates government and policy interventions that prioritize energy access in underdeveloped areas. The overarching objective is to contribute to the broader discourse on sustainable development, particularly in the context of SSA, by demonstrating how improved energy access can serve as a fundamental lever for lifting rural populations out of poverty. In short, the study provides insights into how renewable energy projects can serve as a means of community service, build community cooperation, and serve as a model for future similar initiatives in other rural areas.

The article is structured as follows: Chapter 2 is a detailed description of the research method, including a description of the data and statistical analysis. This chapter also includes a map of the study area. Chapter 3 presents all the results obtained. These findings are discussed in Chapter 4. Chapter 5 presents the final conclusions of the research and future researches.

Methods

Data description

This section is devoted to the justification of the choice of the site, sampling techniques, and data collection. In the former, we provide detailed reasons for choosing the Mbo'o community in the coastal part of Cameroon in general and the village of Nsanke in particular, as the study area. Finally, the sampling techniques and data collection are presented. Here we take the opportunity to provide the reader with the location map of the study area.

Justification of the choice of the site

The choice of Nsanke village for our research stems firstly from the fact that the community faces high levels of poverty. Another reason is that Nsanke village is one of the first ten (10) CASCI villages in the Mbo'o community in the coastal part of Cameroon. The program, initiated by the ASAIE organization with partners including the *Laboratoire Energétique Carnot* (LEC), the Promotion Centre of Research for Technological Advancement and Sustainable Development (PCR-TASD), and PROSOFOR AFRIQUE, aims to fight against poverty and rural exodus and to stimulate the local economy in the Central African region. The key elements of interest are: (1) access to energy; (2) access to improved drinking and affordable water; and (3) availability of food. Access to energy in Cameroon is possible through connection to the central public grid. However, some locations, such as the Mbo'o community in the coastal part of Cameroon, cannot be connected to the public grid due to their location, distance from the central grid, geographical conditions, and road accessibility. In addition, given the challenges faced by the Mbo'o community, it seems technically complex to extend the network to the population. Their lack of connection to the public grid means that they are classified as "remote areas" that can only benefit from decentralized electrification. Aware of these considerations and of the current difficulties of access to energy in SSA and in the remote areas of Cameroon, the ASAIE organization has set up the CASCI program to address their problems of low energy tariffs, in particular by making more productive use of energy and providing them with clean and affordable water and food.

Sampling techniques and data collection

Taking into account the subjectivity of the researchers and the characteristics of the remote rural communities, the present study opted for the probability sampling method, which is "based on a random or chance selection method, so that the probability of selecting elements of the population is known" [49]. The sampling method consisted of non-proportional stratified random sampling. This involves dividing the population into homogeneous groups, called strata, which are mutually exclusive. As the strata must be mutually exclusive, sampling is then applied within the subgroups. This sampling method supported the random selection of Nsanke village. Data were collected using the Paper and Pencil Interviewing (PAPI) method, which is a paper-based method using a semi-closed questionnaire. The door-to-door survey collected data on household demographic and socio-economic characteristics, basic social services, development infrastructure, and lighting. A Garmin handheld GPS (Global Positioning System) receiver was used to locate the study area (see Fig.1). A questionnaire survey was conducted in Nsanke village from April to May 2023. At the beginning of the study, 100 households were selected. However, due to constraints encountered in the field, including but not limited to fear and unavailability of the population due to agricultural activities, 68 were interviewed and were designed to be representative at the Nsanke village level. In order to dispel the doubts about the handling of missing data in relation to the reduction of the initial sample size from 100 to 68 households, given the constraints mentioned above, we decided to proceed with field observations, as the information provided by respondents regarding the purpose of this study was almost identical. Thus, much of the other information collected was supplemented by field observations.

Statistical analysis framework

In this section of the research methodology, we describe in detail how the data collected in the field were analyzed. We have focused our attention firstly on the descriptive analysis and secondly on the BLR within the framework of the multivariate explanatory statistical analysis. The former allows us to look at the main characteristics of the population, while the latter allows us to examine the relationship between energy deprivation and poverty in Nsanke and neighboring villages. Apart from descriptive analysis, which provides a detailed

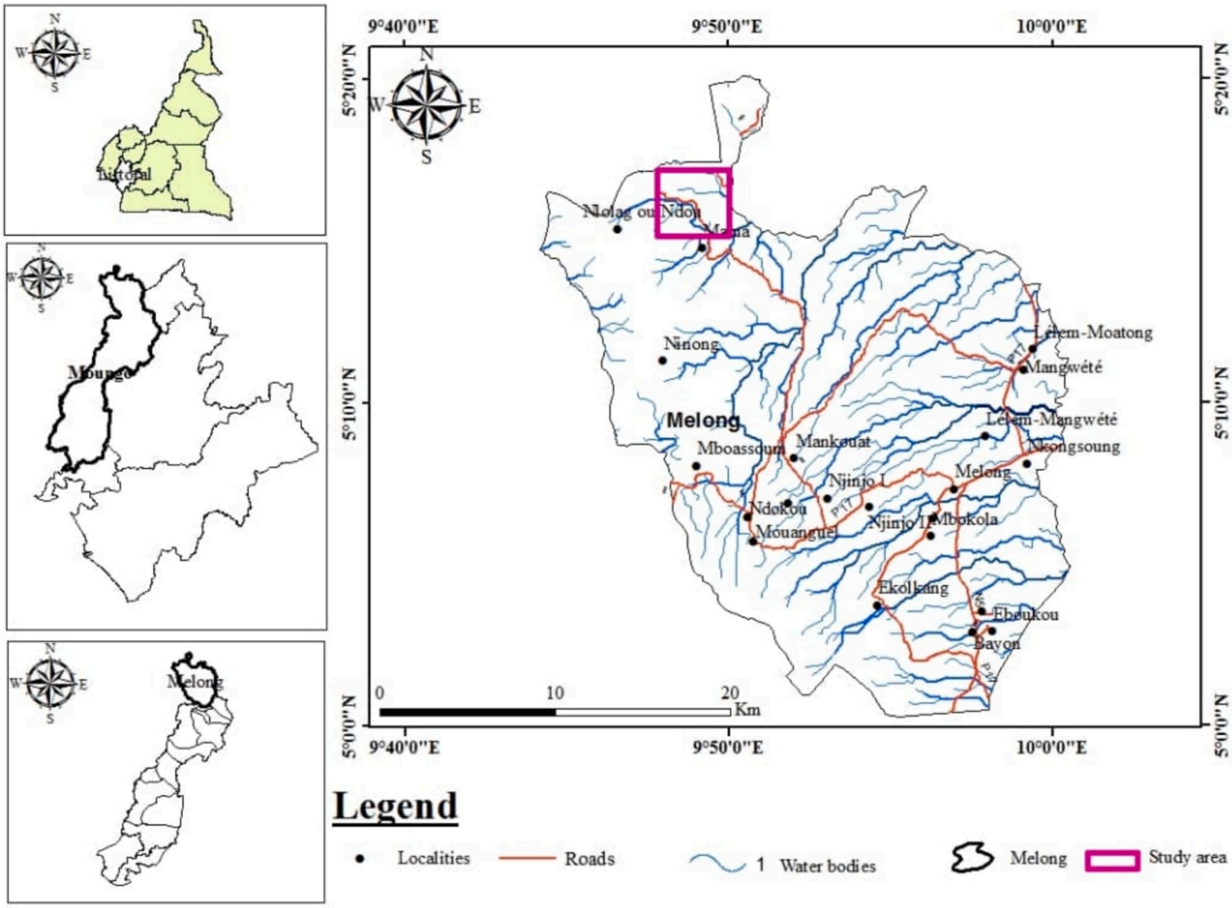

Fig. 1. Location of the study area. On the right, the pink frame represents the study area, which includes Nsanke village and neighboring villages such as Njinjou, Mankwa, Ebang-Mama, Mama, Etabang, Mbokambo, Ekah, Ebakong and Nlolak. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

description of the variables used and the direct relationships between them, the choice of multivariate analysis method depends on the nature and form of the dependent variable(s). For example, if the analysis method is quantitative, as is the case in this study, the OLS (Multiple Linear Regression) method is recommended. For instance, the choice of BLR is due to the fact that our variables are binary or categorical. Another reason for choosing this method over others is that it takes into account all the variables used in the model. Several authors [9,48] have found this model to be a useful tool for investigating factors influencing poverty with dichotomous categorical response variables.

Data processing

Mapsource software version 8.5 was used to import data from the handheld GPS receiver into the machine. Data processing was carried out using SPSS version 23 software (Statistical Package for Social Sciences). These data were then converted into vector data to be imported into Arcgis software version 10.5, which allowed the creation of the localization map developed in this study (Fig.1).

Model Specifications

In this paper, we have examined both descriptive (flat sorting) and multivariate explanatory statistical (BLR) analysis. The former is used to assess the extent of poverty, specifically energy poverty. It describes the poverty profile of the population surveyed and finds that it manifests itself in socio-economic characteristics, mainly in the difficulty of access of the population to basic social services, such as access to drinking water and energy services. The latter is used, for example, to examine the relationship between poverty and energy. The BLR model, which is used as the dependent variable (use of incandescent lamps as the only means of lighting), is coded as 0 or 1 for two possible categories.

The dependent variable describing alternative energy use is a dichotomous variable with only two levels. Households with kerosene lamps as the only means of lighting were binary coded (1). All other households using other means of lighting, such as solar light, generators, candles, and solar torches, were coded (0). There are 6 independent poverty variables used in this study and coded according to binary logic. In the second Cameroon Household Survey (CHS2), the National Institute of Statistics [50] establishes the poverty profile in Cameroon by describing the living conditions of Cameroonians. Five variables related to the head of the household were considered relevant to characterize poverty. These are: gender, level of education, age, activity, and household size. According to the CHS2 data, we focused our analysis on poverty using socio-economic variables of households (age, sex, activity, education, size, and consumption). The description of the model variables chosen to estimate the relationships outlined is defined in Table 1 below.

Our hypothesis is that there is a relationship between the socio-economic indicator of poverty and a household's energy indicators. We consider that our sample consists of individual independent variables indexed. For each individual in the sample, we observe whether a particular event has occurred, and we note the coded dependent variable associated with the event. We record whether the event occurred for the individual and whether it did not. The theoretical formulation of such a model is expressed in Eq. (1) [51].

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \ldots + \beta_n X_n + \varepsilon \qquad (1)$$

Table 1
Description of model variables.

Variable	Variable type	Description
Dependent variable		
Traditional sources of energy used	**Binary**	1 = use of kerosene lamp as the sole means of lighting 0= No use of kerosene lamp as the sole means of lighting
Independent variables		
Sex of head of household	**Binary**	1 = Female 2 = male
Age of head of household		1 = less than or equal to 35 years 2 = over 35 years
Household size		1 = less than or equal to 6 peoples 2 = More than 6 peoples
Level of education of household		1 = less than or equal to Primary level 2 = over than or equal to Secondary level
Main activity of head of household		1 = Agriculture 2= Other
Monthly consumption of household		1 = Less than or equal to $25 2 = More than $25

The logit transformation of Eq. (1) is given by log-odds in Eq. (2).

$$Logit(Y) = Ln\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \cdots + \beta_n X_n + \varepsilon \tag{2}$$

Then the odds ratio index is given in Eq. (3), where P is given in Eq. (4). P is the probability that a household uses kerosene lamp as the only alternative means of lighting. The set β_i contains the estimates of the regression coefficients or their approximate contribution. A positive coefficient indicates an increase and a negative one a decrease in the probability of a household using the household energy indicators. ε is the error term, which is a vector of all unobserved variables and follows a normal distribution for all independent observations.

$$Exp(\beta) = \frac{P}{1-P} \tag{3}$$

$$P(Y=1) = \frac{e^{\beta_0 + \sum_{i=1}^{n} \beta_i X_i}}{1 + e^{\beta_0 + \sum_{i=1}^{n} \beta_i X_i}} \tag{4}$$

Results

Descriptive analysis

This section describes and summarizes the main characteristics of our data. These characteristics are essentially the demographic and economic characteristics of the study population in Nsanke village, the lighting medium used by the population and its accessibility, the status of road infrastructure in Nsanke village and adjacent areas, and access to potable and affordable water. In the same vein, we statistically analyze the interest of the population in renewable energy resources, providing the distribution of the population according to renewable energy lighting medium.

Demographic and economic characteristics

The results of the data on demographic and economic characteristics are presented in Table 2. The table shows that the rural population is predominantly male (72.58 %). Female household heads represent only 27.42 % of the survey population. 64.52 % of household heads are over 35 years old and are mainly farmers (93.55 %). The distribution of the sample by household size shows that the families in our sample have an average of 7 people, but there are some rare households with up to 10 members. In order to meet their needs, somebody invests in small, non-agricultural businesses in rural areas. The surveyed population sample consists mainly of people with secondary education. Up to 54.84 % of the heads of households have completed secondary education. The purchasing power of the population is quite low. 91.93 % of households have a total monthly consumption that does not exceed $25.

Table 2
Demographic and economic characteristics.

Major characteristics of the population in Nsanke village	Effective	Frequency (%)	Cumulative Frequency (%)
Gender			
Female	17	27.42	27.42
Male	45	72.58	100.0
Total	62	100.0	
Age			
Less than or equal to 35 years	22	35.48	35.48
Over 35 years	40	64.52	100.0
Total	62	100.0	
Level of education			
Less than or equal to Primary level	28	45.16	45.16
Over than or equal to Secondary level	34	54.84	100.0
Total	62	100.0	
Main activity of household			
Agriculture	58	93.55	93.55
Other	4	6.45	100.0
Total	62	100.0	
Household size			
Less than or equal to 6 peoples	46	74.19	74.19
More than 6 peoples	16	25.81	100.0
Total	62	100.0	
Monthly expenditure in total household consumption			
Less than or equal to $25	57	91.93	91.93
More than $25	5	8.07	100.0
Total	62	100.0	

Energy resources found in Nsanke and neighboring villages

Tables 3 and 4 show the lighting media found in Nsanke and neighboring villages. A graphical representation of these values is shown in Fig. 2. Fig. 2 shows that kerosene lamps are the most commonly used lighting medium by the surveyed population (Fig. 3a). 79.03 % of the households use kerosene lamps as the only means of lighting. The other energy sources, including generators (Fig. 3b), battery-powered torches (Fig. 3c), candles, and solar panels, are not used systematically but alternately or concurrently with kerosene lamps. The figure (Fig. 2) shows that photovoltaic energy is the only renewable energy source found in Nsanke village, and it is generally used at a low rate (7 % for solar torches, 4 % for solar lamps, and 2 % for solar panels). In Nsanke village, as well as in neighboring villages such as Mama, Ebang-Mama, Etabang, Mbokambo, Ekah, Ebakong and Nlolak, to mention just a few, cooking is essentially done with firewood (Fig. 4a) as the primary resource, under a three-stone hearth [Fig. 4(b,c)] as a convection element for all the inhabitants (100 %) of the village.

Table 3
Distribution of the sample according to the lighting medium used.

Lighting mediums used	Effective (%)	Cumulative frequency (%)
Kerosene lamp	79.03	79.03
Other lighting mediums	20.97	100.0
Total	**100.0**	

Table 4
Distribution of the sample according to renewable energy source used.

Electrification mean	Solar Torch	Solar Lamps	Solar panels
Frequency Yes	20	12	5
No	42	50	57
Total	62	62	62
Access rate	32.3 %	19.35 %	8.06 %

Road and clean water accessibility

Fig. 5 shows the poor road access in the study area. Nsanke village is isolated, and the access roads are tracks or dirt roads that are impassable to users during the rainy season [Fig. 5 (a,b)]. Fig. 5c shows some of the water sources found in the village. Almost all Nsanke villagers do not have access to clean and affordable water for drinking and cooking. In order to provide the population with clean and affordable water, the poorest water sources can be arranged. These water sources include rivers (Fig. 5c) and wells.

Multivariate explanatory statistical analysis of the relationship between lack of access to energy and poverty of inhabitants in the Nsanke village

Table 5 summarizes the variables used in the model in Eq. (3) to assess the relationship between access to energy and poverty. With regard to the explanatory variables for lack of access to energy, each of them has an internal probability level of 5 %, with positive and significant odds ratio indices (Exp(β)). This confirms that each of them has a significant impact on the explained variable. Gender (0.252), age (0.263), education of the head of the household (0.034), and household size (1.120) have a positive effect on the probability of respondents using kerosene lamps as the only means of lighting in the sample. Moreover, the effect is negative for the activity of the head of the household (−1.250) and the consumption of the household (−0.522). We can see that the variables representing unspecified factors (constant) are positive and significant at the 1 % threshold, which expresses that socio-economic factors as determinants of poverty can predict the lack of access to energy. In addition, the Chi-square statistic, which confirms the specification of the model, is significant at the 5 % level. There is a positive and significant relationship between lack of access to energy and household poverty. Specifically, the poverty level of households in Nsanke village is perfectly related to the lack of access to energy. According to Nagelkerke's R2, we conclude that the model explains with a significance rate of 4.7 % that the lack of access to energy of the respondents in Nsanke village influences the poverty indicators. Nagelkerke's R2 or pseudo-R2 shows the goodness of fit of the model; its low value shows that other factors not captured in our model influence the

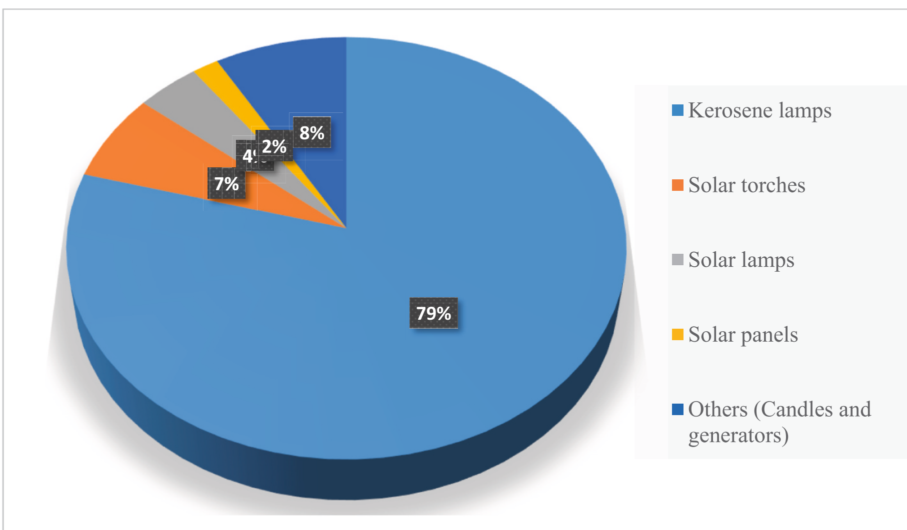

Fig. 2. Energy resources encountered in Nsanke and neighboring villages.

Fig. 3. The lighting medium used by the surveyed population in Nsanke village and neighboring villages. (a) Kerosene lamp; (b) Generator; (c) Battery-powered torch.

Fig. 4. Essential means of cooking used in Nsanke village as other villages in the Mbo'o community in the coastal part of Cameroon. (a) Wood storage for future food cooking use; (b) Use of wood by women for food cooking; (c) Use of wood by men for palm oil production.

Fig. 5. Poor road access and lack of clean and affordable water in Nsanke village. (a and b) Impractical roads leading to Nsanke village and neighboring villages; (c) River found in Nsanke village and used by the population for drinking and cooking.

Table 5
Binary Logistic Regression Coefficients Matrix Testing the Effect of Poverty on Lack of Access to Energy.

Poverty indicator	β	E.S	Wald	ddl	Sig.	Exp (β)
Sex(1)	0.252	0.192	1.712	1	0.001	1.286
Age(1)	0.263	0.194	1.828	1	0.006	1.300
Size(1)	1.120	0.414	7.330	1	0.007	3.066
Education(1)	0.034	0.251	0.018	1	0.002	1.035
Activity(1)	−1.250	0.295	17.933	1	0.000	0.287
Consumption(1)	−0.522	0.260	4.027	1	0.045	0.593
Constant	2.835	0.743	14.543	1	0.000	17.027

R-2 of Nagelkerke = 0.047 −2log-likelihood = 854.565[a]
R-2 of Cox & Snell = 0.030 Khi-square = 3.324** P-value = 0.04
**: Significant at 5 % consecutively
a. Introduction of variables in step 1

choice of energy used, but it does not have any effect on the model because the dependent variable in logistic regression is not continuous.

Discussions and recommendations

Discussions

Both descriptive and multivariate explanatory statistical analysis of the survey data allowed us to assess the extent of poverty and the level of penetration of renewable energy in Nsanke and neighboring villages. Lack of access to energy is characterized by a strong dependence on traditional fuels (100 % firewood, 79.03 % kerosene), a lack of energy transition behavior (0 % use of LPG) and, finally, a low rate (32.3 % for solar torches, 19.35 % for solar lamps, and 8.06 % for solar panels) in the use of renewable energies. This is due to the low purchasing power of the population and the high costs associated with the use of kerosene lamps or solar panels and other equipment related to the production of solar energy (battery, inverter, solar panels, and solar torches). These main characteristics of the population in Nsanke and neighboring villages due to lack of access to energy have been observed by the authors of Refs [52–54] for strong dependence on traditional fuels, by the authors of Refs [55] for energy transition behavior, and by the authors of Ref [56] for low rate in the use of renewable energy sources and devices. However important it is for a community or a nation as a whole to move towards a more sustainable and energy-efficient future by developing sustainable energy sources, addressing energy security issues, and supporting international efforts to tackle climate change, in Nsanke and neighboring villages, as in many rural areas around the world, access to reliable and affordable energy remains a major challenge [56]. According to [56,57], this is due to a number of factors, including underdeveloped energy infrastructure, long distances from major energy production centres, and a lack of investment in the rural energy sector. According to [55], the use of the above-mentioned traditional energy sources in Nsanke and neighboring villages can lead to indoor air pollution, which can cause respiratory problems and other illnesses in the population. In addition, the time and energy required to collect traditional fuels places an additional burden on the rural population, especially women and children, reducing the time that could be spent on education or other productive activities [55].

The profile of the population surveyed is thus similar to the typical profile of the poor worldwide (individuals living in rural areas, young people, the illiterate, farmers, and members of large families with several children) [58], which represents a major obstacle to the development of energy access for these populations. Such a result was also obtained by Kenfack et al. [59], who found that the reduction of poverty in remote rural areas in Africa, and particularly in Cameroon, inevitably requires the removal of unusual obstacles as well as the development of sustainable energy projects that address (i) literacy, (2i) poor roads, (3i) means of transport, and improved data systems to track the progress of electrification.

In terms of the relationship between energy and poverty, female-headed households are 1.286 times more likely than male-headed households to use kerosene lamps as their sole means of lighting. These findings are consistent with other studies [17, 46] showing that a male head of household is more likely to use clean fuels than a female head of household. The probability of using kerosene lamps as the sole means of lighting is 1.300 times higher for heads of households under 35 years of age than for heads of households over 35 years of age. These findings are consistent with Tchereni's studies [61], which show that there is a negative relationship between dirty fuel and age. It can be seen that households with less than or equal to 6 persons are more likely to use kerosene lamps as the only means of lighting than households with more than 6 persons, with an odds index of 3.066. This contradicts the findings of Deshmukh (2014) [60], who concluded that large households use non-modern fuels and smaller households have a high probability of using modern fuels. This is due to the fact that large households use multiple sources to meet their energy needs. The probability of using kerosene lamps as the sole means of lighting is 1.035 times higher for heads of household with any formal education or at least primary education than for heads of household with more than or equal to secondary education. The results are in line with other studies [62,63] showing that the education level of the household head is positively related to the use of clean fuels. A one-unit increase in agriculture as the main activity of the household would lead to a 0.29 decrease in the probability of using kerosene lamps as the sole means of lighting in favour of using more other energy sources. This is consistent with research by Link and Ghimire [64], who suggest that households without employment lead to greater reliance on traditional forms of energy. A one-unit increase in household consumption would reduce the probability of using kerosene lamps as the sole means of lighting by 0.59 in favor of using more modern energy sources. The low level of consumption currently forces people to use traditional energy sources. This result confirms the results obtained by the authors of Refs [65–67], who explained that as income increases, the use of traditional energy sources decreases, leading to an increase in the use of modern energy sources. The study shows that in the village of Nsanke, as in remote rural areas, large families whose heads are young, female, farmers, and illiterate, with low monthly consumption, are likely to use several alternative sources for their lighting needs in addition to kerosene lamps. This result is in line with the findings of Ogwumike et al. [17], Okwanya et al. [68], and Desalu et al. [69] who found that the likelihood of being poor reduces the likelihood of using modern and clean energy sources. The result is also similar to that of Kamdem [70], who showed that poverty reduction through effective access to electricity must be achieved by improving certain poverty indicators in rural electrification.

Recommendations

From the above findings and discussions, this study agrees with the existing literature on certain key conclusions. Lack of access to energy as the main cause of poverty for the inhabitants of Nsanke and neighboring villages has been observed in other regions of the world, which facilitates the proposal of solutions to overcome energy poverty in the study area. Our recommendations provide specific policy advice not only for alleviating energy poverty in Nsanke and neighboring villages but also for addressing energy poverty in other rural areas of Central Africa and around the world. In order to improve the living conditions of the inhabitants, we suggest that the government should pay more attention to Nsanke and neighboring villages, whose inhabitants are currently overlooked in development initiatives. Our suggestion in the action that should be taken to improve the living conditions of the people of Nsanke is to implement rural electrification projects, as Nsanke and the neighboring villages have natural advantages for clean energy with abundant solar (Fig. 6a), wind (Fig. 6b), and hydropower reserves. The benefits of implementing such projects, which will lead to the development of renewable energy sources such as solar [71], wind [72], and micro-hydropower [73], include protecting the environment, especially in a world where fossil fuels are becoming scarce and climate change is a reality [74], improving individual quality of life, facilitating community services such as health and education, and creating small economic activities that generate income for the population. In fact, authors of ref.[75] revealed that if the population in SSA had other means of energy access, they would be very productive people. At the same time, financial support and other approaches can be used to help the population and pastoralists accelerate the upgrading of energy-using facilities while increasing the population's capacity to use clean energy. In this way, the government can take on the responsibility of mass-producing the solar lamps designed and manufactured in our recent studies [44] and distributing them at affordable prices tailored to their income levels, as these lighting devices have proven to be the most practical solution for improving the general standard of living of the population in the Mbo'o community.

For instance, the practical implementation of these rural electrification projects in Nsanke and neighboring villages requires not only financial resources but also the active participation of communities and the government in the decision-making process, as well as an in-depth analysis of the political and regulatory barriers that may hinder the implementation of these projects and the proposal of possible strategies to mitigate them. All these important aspects of the successful implementation of rural electrification projects in Nsanke and neighboring villages constitute our future research objectives, as mentioned in the conclusion below.

Conclusions

The purpose of this research study was to analyze the contribution of energy to poverty alleviation in Nsanke and neighboring villages using both descriptive and BLR tests. Descriptive analysis allows us to look at the main characteristics of the population, and BLR allows us to examine the relationship between lack of energy and poverty in Nsanke and neighboring villages. The data used were collected in the field using the PAPI method with a semi-closed questionnaire. The descriptive analysis of the data collected in the field has shown that the energy situation in Nsanke and neighboring villages is characterized by a strong dependence on traditional fuels and a low percentage of use of renewable energy. Meanwhile, the BLR shows a strong correlation between access to energy and poverty alleviation. This means that the people of Nsanke are predominantly poor because they lack access to energy, especially electricity. The present study, which represents one of the first research work of CASCI in the Mbo'o community in the coastal part of Cameroon, stands as a road towards many other research works for the proper implementation of sustainable development in Nsanke, considered as the first village of CASCI and neighboring villages including Njinjou, Mankwa, Ebang-Mama, Mama, Etabang, Mbokambo, Ekah, Ebakong and Nlolak. In the future, we therefore plan to carry out the following research for the practical implementation of rural electrification projects in Nsanke and neighbouring villages: (i) Social and economic impacts of lack of energy in the Mbo'o community: The case study of Nsanke and

Fig. 6. Nsanke and the neighboring villages natural advantages for the development of renewable energy sources. (a) Solar radiations, (b) Wind.

neighboring villages; (2i) Environmental impacts of implementing rural electrification projects in Nsanke and neighboring villages; (3i) Study, design, and installation of photovoltaic solar street lamps for public lighting in Nsanke village; (4i) Measurement and monitoring of energy poverty alleviation in Nsanke and neighboring villages, and (5i) Policy and regulatory barriers to the implementation of rural electrification projects in Nsanke and neighboring villages: potential policy-based approaches.

Author contributions

All the authors contribute as indicated in the credit authorship contribution statement below.

CRediT authorship contribution statement

C.M. Ekengoue: Methodology, Conceptualization, Writing – original draft, Writing – review & editing, Investigation, Validation, Formal analysis, Project administration. **J.T. Diffo:** Supervision, Writing – review & editing, Writing – original draft, Formal analysis. **V. de Dieu Bokoyo:** Conceptualization, Writing – original draft. **M.D. Fendji:** Writing – original draft. **V. de Paul Fotso:** Funding acquisition. **C. Kenfack-Sadem:** Methodology. **L. Akana Nguimdo:** Supervision. **G. Ngoukwa:** Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All data generated or analyzed during this study are included in this published article.

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